

SWASTIK CHOWDHURY

ssamadda@alumni.cmu.edu | 412-844-1548 | [LinkedIn](#) | [Github](#) | [Portfolio](#) | [Medium](#)

EDUCATION

Carnegie Mellon University

August 2024 - August 2025

Master of Information Systems

Coursework: Machine Learning, Deep Learning, Machine Learning in Production (TA), Database Management Systems, Distributed Systems, DevSecOps, Data Focused Python (TA), Object-Oriented Programming with Java

MAKAUT

June 2015 - June 2019

Bachelor of Technology

Coursework: Computer Programming, Data Structures & C, Object-Oriented Programming, Database Management

WORK EXPERIENCE

Vertisage Technologies, Data Engineer

April 2023 - March 2024

- Led migration of **80+** legacy Redshift MPP stored procedures to Apache Spark (Scala) ETL on EMR with Airflow orchestration, cutting annual infrastructure cost by **\$1M+** and reducing job runtime by **40%** while maintaining full data correctness.
- Implemented a Debezium-based Change Data Capture pipeline writing to Iceberg tables on S3, delivering sub-**5-minute** updates to fraud analytics datasets consumed by the Risk team.
- Owned primary on-call PagerDuty for **25+** production pipelines serving **4 downstream teams**, authoring runbooks and automating recurring failure modes to cut repeat incidents by **60%** and reduce **MTTR from 45 min to 15 min**.

Tata Consultancy Services, Software Engineer

July 2019 - April 2023

- Optimized Apache Spark (Python) ETL by diagnosing data skew via Spark UI stage metrics and applying salting and broadcast joins cutting runtime by **40%** and cloud compute cost by **15%** across production pipelines.
- Built Apache Spark (Python) ETL pipelines with Airflow to ingest **5+ GB/day** of user-interaction JSON events into S3 from AWS Kinesis, partitioning by date and event type to cut Athena scan volumes and query costs by **40%** and drop analyst query latency from minutes to seconds.
- Designed dimensional data marts using dbt incremental models and Snowflake clustering strategies, improving reporting performance and reducing warehouse compute costs by **20%**
- Enforced strong data quality with AWS Deequ for nulls, distribution drift, anomaly detection, and wrote unit/integration tests for TDD, driving reliability through code reviews.

PROJECTS

LLMOps : Real-Time AI Coaching Assistant | GitHub

- Built a real-time AI coaching app with a **multi-agent architecture** (conversation, coaching, and memory) streaming responses token-by-token over **WebSockets**, backed by a model registry with live rollback.
- Implemented **RAG** long-term memory (ChromaDB + recency-aware reranking) and a hybrid grounding-verification pipeline (NLI + LLM-as-judge) to **reduce hallucinated** coaching citations.
- Engineered safety and privacy controls : crisis-detection gate, HMAC-SHA256 PII redaction.
- **JWT/OAuth** auth with **per-user data isolation** across MongoDB, Postgres, and the vector store.
- Shipped full observability (Prometheus/Grafana), an eval harness, and a Dockerized GitHub Actions CI/CD pipeline publishing to GHCR.

MLOps : Recommendation Systems (SCS CMU) (1M users, 27K movies, <500 ms latency, 99% availability)

- Built real-time data pipelines ingesting from **Kafka** and orchestrating data using Airflow into MySQL.
- Containerized Flask services with Docker, deployed them to **Kubernetes** for load balancing.
- Setup **Jenkins** CI/CD and ensured observability with Grafana and Prometheus.
- Measuring traceability with **MLflow** and also monitoring for data drift using KL divergence.

SKILLS

Programming: Python, Scala, Java, SQL

Databases: MySQL, Postgres, MongoDB, Neo4j, Redis, Redshift, Snowflake, Iceberg, Delta Lake, Presto/Trino

Data Engineering: Spark, Hadoop, Kafka, Airflow, Flink, dbt

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, Jenkins, Prometheus, Grafana, Linux (Bash), Postman, Git